# Gabay — Shipping a Public RAG Product

One-page reflection · Graded Mini Project 14.3 · 2026-10-11 · gabay-fint.vercel.app · github.com/jacobvillar/gabay

## Overview

Gabay is a public, domain-specific RAG assistant that answers questions about **Philippine statutory worker benefits** — 13th-month pay, leave, SSS, PhilHealth, and Pag-IBIG — strictly from official government handbooks, with the source passages shown under every answer. It is built with **Next.js + the Vercel AI SDK** (streaming chat, retrieval exposed as a tool call), **Groq** (open-weight GPT-OSS 120B) for generation, and **Upstash Vector** (built-in embedding model) for retrieval, deployed publicly on Vercel.

## Corpus and key decisions

| Decision | Choice | Why |
|---|---|---|
| Corpus | DOLE Handbook, SSS Citizens Charter, PhilHealth Konsulta circular, Pag-IBIG briefer | Public-domain PH government works — legal to serve publicly; a domain where base LLMs are unreliable, so retrieval + citations add real value |
| Generation | Groq — open-weight GPT-OSS 120B | Free tier, no OpenAI key, supports tool calling, deploys on Vercel |
| Embeddings | Upstash built-in model | Removes the embedding API key entirely — one fewer secret |
| Chunking | Per-page → sentence/paragraph-aware packing to ~1000 chars, 150 overlap | Per-page parsing preserves real **page numbers**; sentence-aware packing keeps rules intact |
| Metadata | `document_title`, `agency`, `page`, `source_url` | Readable citations that disambiguate across four agencies |

## The trickiest decision

Making retrieval **trustworthy on high-stakes questions**. A wrong peso figure or rule is worse than no answer, so retrieval had to be a deliberate **tool call** (not always-prepended context), with a tool description tuned so the model reaches for it on benefit questions but skips small talk, and a system prompt kept strict: answer only from retrieved passages, and say plainly when the handbooks do not cover something. Balancing that faithfulness against still being genuinely helpful was the hardest part to get right.

## What it does well

Grounded, cited answers on PH-specific rules. "How is 13th-month pay computed?" triggers the retrieval tool, answers from the DOLE handbook (1/12 of basic salary, with the correct exclusions), and renders the exact source pages — something a raw model gets subtly wrong with no citations.

## What it does badly

Retrieval balance. Because the SSS charter dominates the index, some non-SSS queries can surface less-relevant SSS passages; structure-aware chunking keeps each passage coherent, so per-agency balance is the main remaining lever. These are quality, not correctness, issues — but they are real.

## What I'd do with another week

1. **Rebalance the corpus** — trim the SSS charter to its benefits sections or add per-agency retrieval filters.
2. **Hybrid retrieval + reranking** — combine dense search with keyword matching and a reranker.
3. **Eval harness** — score faithfulness and context-relevance on the demo-question set.
4. **Inline citation mapping** — true [1]/[2] markers tied to the sources panel.